Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Average acceleration

Meaning in my words How would you explain average acceleration to a friend?	**How can you tell?** Which two velocity readings would you subtract, and in what order?
Example or sketch A cart changes from −2 to +4 m/s in 3 s. What is its average acceleration?	**Non-example and why** A partner divides the whole distance by time. What quantity did that give?

Speeding up

Meaning in my words How would you explain speeding up to a friend?	**How can you tell?** A cart changes from −2 to −5 m/s. What happens to its speed?
Example or sketch Draw three equal-time dots for a cart speeding up to the right.	**Non-example and why** A cart moves right with acceleration to the left. Is it speeding up before it turns?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Slowing down

Meaning in my words How would you explain slowing down to a friend?	**How can you tell?** Velocity changes from −5 to −2 m/s. What happens to the speed?
Example or sketch Draw velocity and acceleration arrows for a bike braking while moving right.	**Non-example and why** “Negative acceleration always means slowing down.” What happens if velocity is negative too?

Reversal

Meaning in my words How would you explain reversal to a friend?	**How can you tell?** On a velocity–time graph, how can you tell a cart actually reversed?
Example or sketch A cart goes from +2 m/s through zero to −1 m/s. Draw its direction before and after.	**Non-example and why** A bus stops, then keeps going east. Does stopping prove it reversed?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Before

After

My marks mean

On assignment: Average acceleration

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two signed XJ velocities are −2 and −6 km/s, two seconds apart. The control room must choose its speed alert.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Telemetry engineer**

Send the average acceleration with its sign and unit. Sketch before and after velocity arrows so the shift lead can confirm the alert.

☐ **Replay commentator**

Narrate what changed over two seconds. Have your producer show which arrow proves whether XJ sped up or slowed down.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Explain why two readings give an average, not the acceleration at every instant.